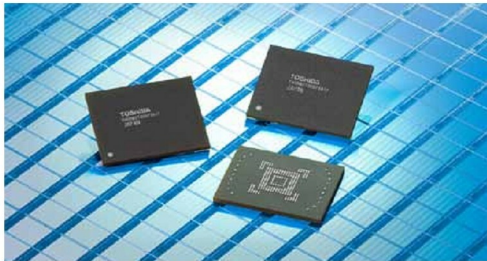


Open NAND Flash Interface (ONFI) Working Group

The Open NAND Flash Interface is an industry working group made up of more than 80 companies that build, design-in or enable NAND flash memory. The group is dedicated to simplifying NAND flash integration into consumer electronics products, computing platforms and any other application that requires solidstate mass storage. They define standardised component-level interface specifications as well as connector and module form factor specifications for NAND flash.

The ONFI Working Group was formed in May 2006 and currently has more than a hundred member companies. ONFI's founding companies include Hynix Semiconductor, Intel, Micron, Phison, SanDisk, Sony and Spansion.



Toshiba's 128GB embedded NAND flash memory module

ness unit, says, "In these cases, portions of the NAND Flash are shadowed into the DRAM and the code executed out of the DRAM and not the NAND Flash. Standard NAND Flash does not provide fast random-access read that is needed for code execution. However, NOR Flash is still used in most mobile phone designs as the code-execution Flash for the baseband processor."

Another interesting trend is that many microcontrollers and even some field-programmable arrays (FPGAs) are now made with embedded flash memory to decrease the component count and cost of embedded systems. All the microcontrollers use an internal, on-chip flash memory for data storage. Embedded flash platforms are also being used for programmable device and systems-on-chip development.

Richard also notes, "While many microcontroller solutions have embedded Flash, the memory density in many cases is not large enough to store

the entire system firmware, especially as application feature sets increase and user interfaces become more interactive and sophisticated. Therefore systems will still incorporate an external Flash solution for direct code execution or for code shadowing."

Write/erase speed on flash memories

Writing to flash memories is a slow process. Harry Kitt, R&D technical associate, Spacelabs Healthcare, shares, "When the system needs to store substantial data, the real-time element is suspended for the system's first purpose. When you are considering the dump-truck-sized flash chip, keep in mind that the only way to fill it is with a teaspoon."

Explaining how the write/erase operation works, Rosenboim says, "Typically, the write speed of flash memories is of the order of a few microseconds per byte, but a write operation will only change bits from '1' to

'0'. So to write arbitrary data, an entire 'block' must be erased first, which will turn all bits to '1'."

It is a major limitation of flash memories that these can read or program a byte or a word at a time, but can be erased only in blocks. So the rewrite and erase operations are not in random-access fashion. An erase operation takes 0.1 to 5 seconds, depending on the specific technology, and typically erases 64 to 128 kB at a time.

Some vendors offer a type of Flash memory that helps alleviate the drawback of spending hundreds or thousands of milliseconds to erase large blocks of memory by allowing the system to erase very small pages of memory. Explaining further, Richard says, "You can erase as little as 256 bytes of memory at a time, and the erase typically takes less than 15 milliseconds. This kind of erase granularity provides the added benefit of not forcing the system to erase more memory than is actually needed for a particular operation, which allows the system to optimise the memory array usage and the system performance."

Flash components primarily come with either of three interface types: Parallel, byte-serial and bit-serial. Rosenboim says, "Since the NOR flash has traditionally been used to replace the EPROM and 'eXecute In Place' the code, it has a parallel address/data interface that is 'memory mapped' onto any type of processor by directly interfacing its local bus signals."

Today, the price of the NOR flash has come down to a point where its package is the most significant component of the cost. To reduce the cost further, a reduction in the number of pins on the package resulted in introduction of 'bit-serial' interface components like serial peripheral interface (SPI) based flash. These parts require a special module in the microprocessor to initialise its DRAM controller and copy the entire boot code section into the DRAM for execution.

"Typically, NAND flash components come with a byte-serial interface,